WEEKLY REPORT 2

Computer Vision Project

Global Wheat Full Semantic Segmentation Using Deep Learning Methods

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1 Introduction

During Week 2, the focus was on understanding the dataset structure, preparing image and annotation data, and performing preliminary preprocessing required for semantic segmentation model training.

2 Work Done This Week

The following tasks were completed during Week 2:

- Downloaded and organized the Global Wheat Head Dataset for experimentation.
- Examined the dataset structure including image files, annotation formats, and metadata.
- Studied the segmentation annotation format used for identifying wheat spikes, leaves, and stems.
- Visualized sample images along with their corresponding segmentation masks to better understand the labeling structure.
- Set up the deep learning development environment using Python and relevant libraries such as PyTorch, OpenCV, and MMSegmentation.
- Implemented preliminary preprocessing steps including image resizing and normalization.
- Prepared dataset directories for training, validation, and testing.

3 Dataset Preparation

Proper dataset preparation is essential for training deep learning models effectively.

3.1 Dataset Structure

The Global Wheat Head Dataset contains high-resolution images of wheat crops collected from different geographic locations and growth conditions. Each image is accompanied by annotations identifying wheat heads and plant structures.

3.2 Image Preprocessing

Images were resized to a consistent resolution to ensure uniform input dimensions for the segmentation model. Normalization techniques were also applied to standardize pixel intensity values.

3.3 Annotation Handling

Segmentation masks corresponding to each image were examined and aligned with the image data. These masks represent pixel-level labels indicating different plant components such as spikes, leaves, and stems.

3.4 Dataset Splitting

The dataset was organized into training, validation, and testing subsets to ensure reliable model evaluation.

4 Initial Visualization

Visualization of the dataset was performed to better understand image characteristics and annotation quality. The following observations were made:

- Wheat spikes often overlap with leaves and stems, making segmentation challenging.
- Variations in lighting conditions and backgrounds exist across images.
- Differences in plant growth stages influence the appearance of wheat structures.
- The dataset contains images from diverse geographic locations, improving model generalization.

5 Work to be Done Next Week

The following tasks are planned for Week 3:

- Implement semantic segmentation models such as U-Net or DeepLab.
- Train the segmentation model using the prepared dataset.
- Evaluate model performance using metrics such as Mean Intersection over Union (mIoU).
- Optimize model performance using hyperparameter tuning.
- Begin integrating segmentation outputs with crop yield prediction analysis.